Lost in the Middle How Language Models Use Long Contexts (2307.03172v3)

Datasets used

1. NaturalQuestions-Open

Abstract

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.1

> While recent language models have the ability to take long contexts as input, relatively little is known about how well they use longer context.

Study to understand how recent language models use their context.

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.1

> We analyze the performance of language models on two tasks that require identifying relevant information in their input contexts: multi-document question answering and key-value retrieval.

Models must use the information in their local context to perform multi-document QA and key-value retrieval

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.1

> In particular, we observe that performance is often highest when relevant information occurs at the beginning or end of the input context, and significantly degrades when models must access relevant information in the middle of long contexts, even for explicitly long-context models. Our

Performance peaks when relevant information is at the beginning of the context. Performance drops when the information is in the middle.

Conclusion

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.11

> We empirically study how language models use long input contexts via a series of controlled experiments. We show that language model performance degrades significantly when changing the position of relevant information, indicating that models struggle to robustly access and use information in long input contexts

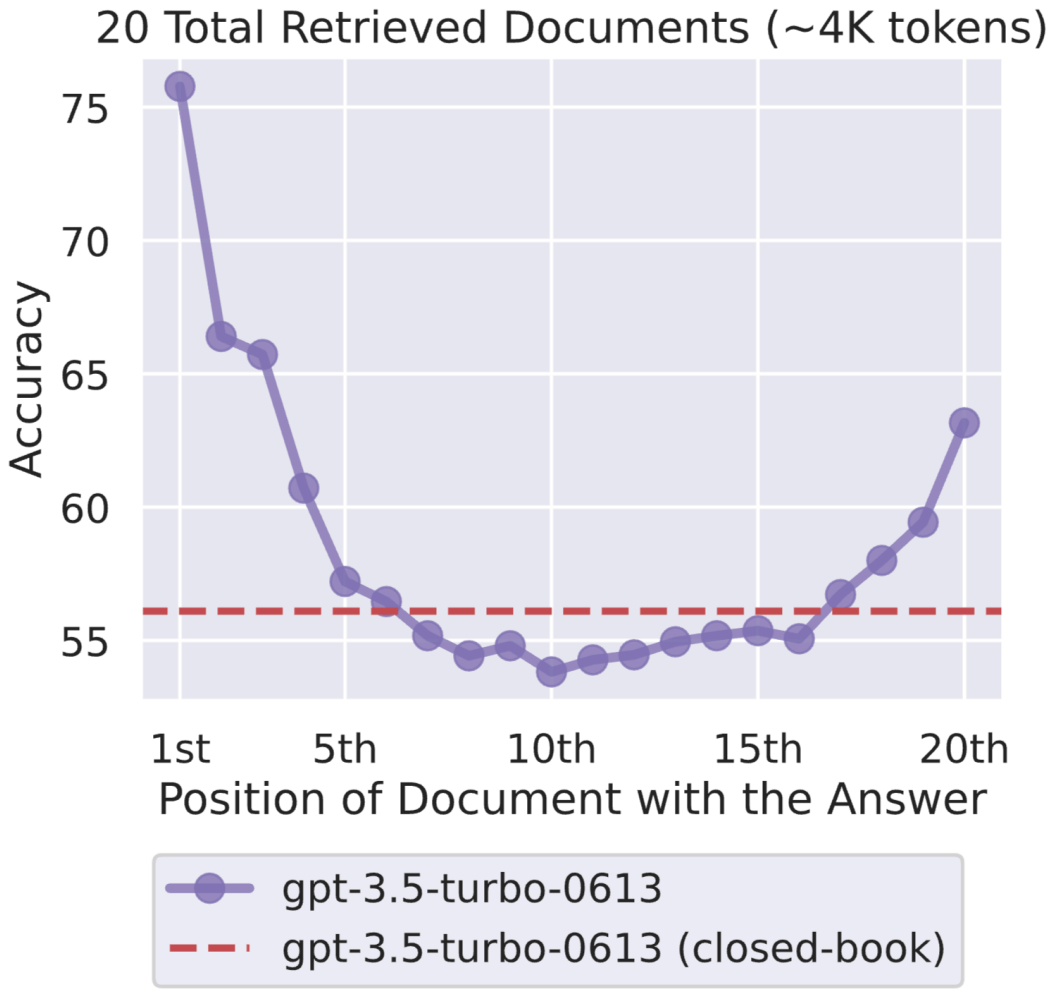
Performance drops significantly when models must access information in the middle of the context.

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.11

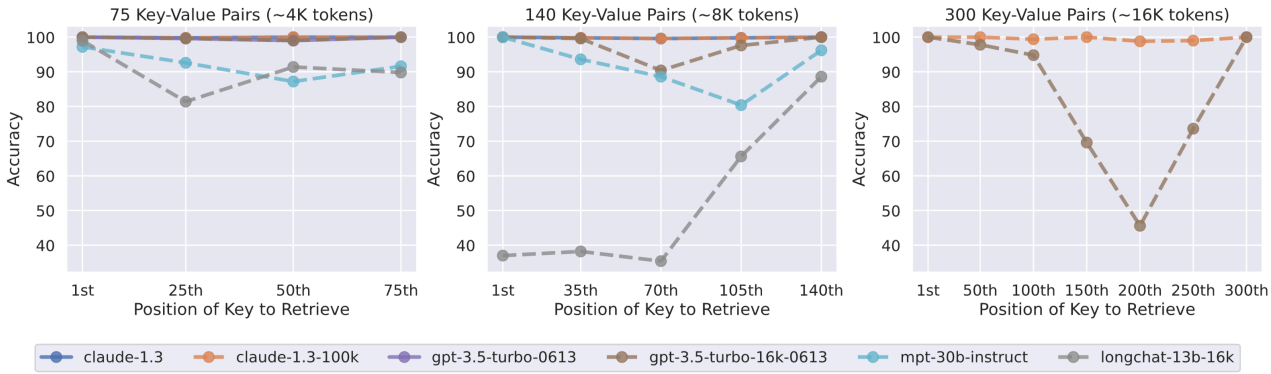
> Finally, we conclude with a practical case study of open-domain question answering, finding that the performance of language model readers saturates far before retriever recall

Implies, even though a new document is added that contains the answer, the models find it difficult to retrieve that information.

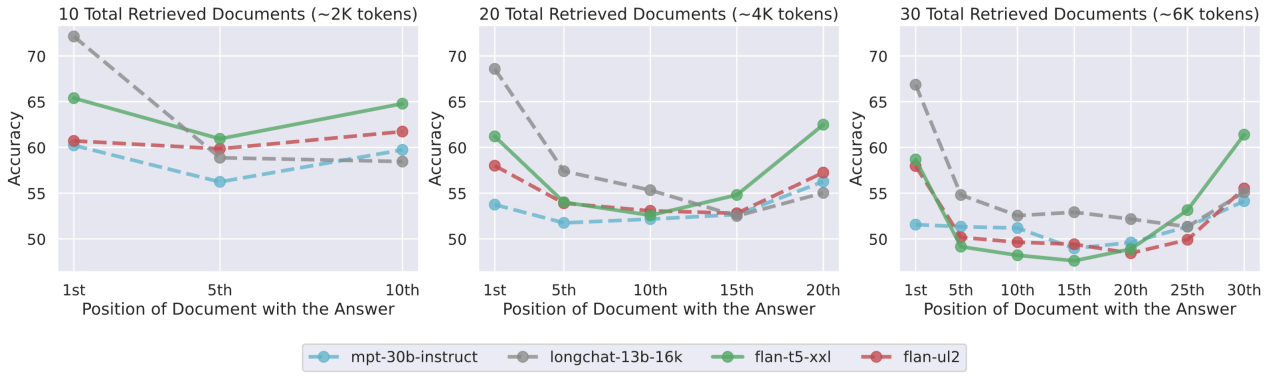
Tables and figures



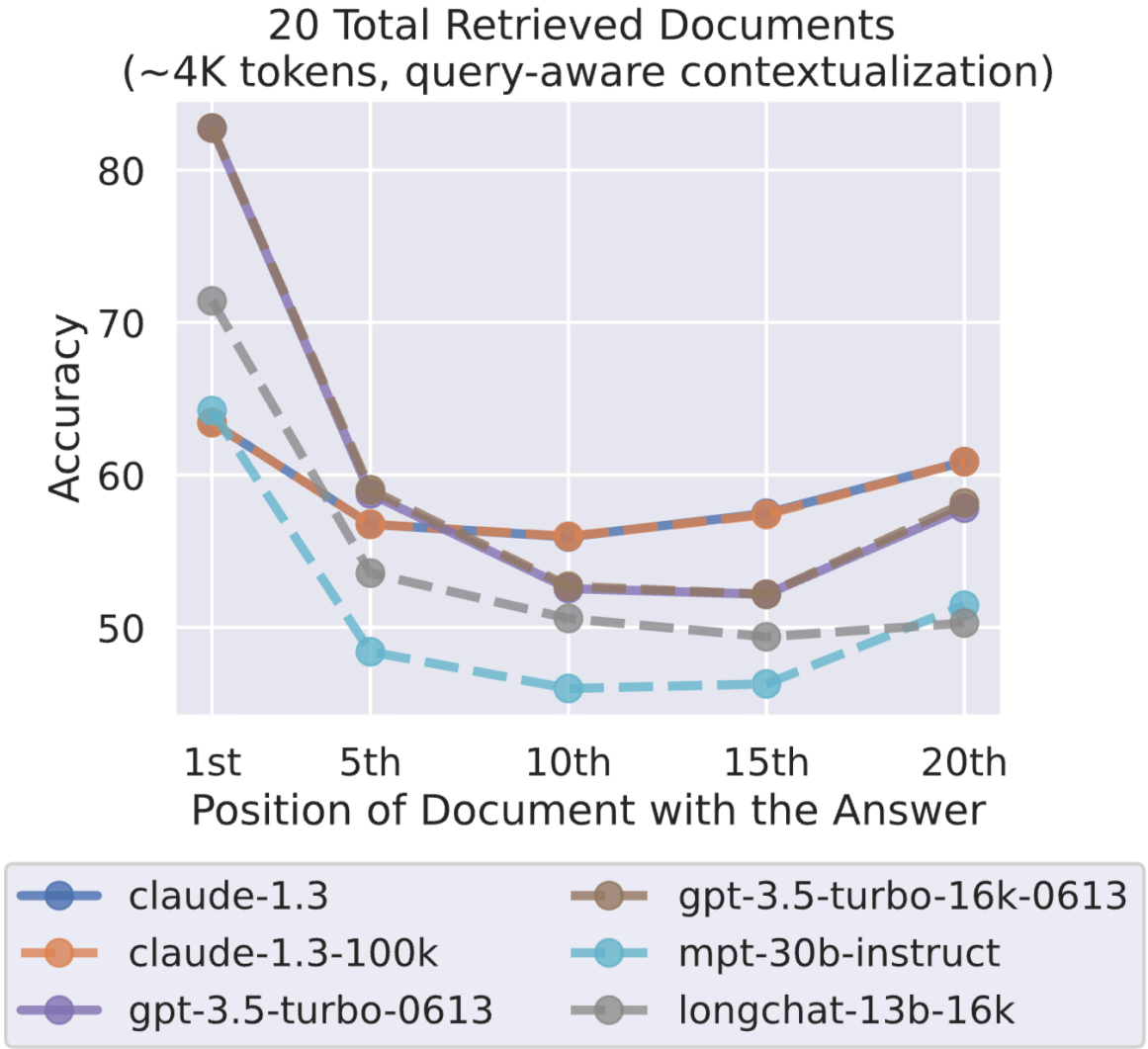
U-shaped curve obtained according to the position of the document containing the answer.



For the synthetic key-value retrieval task, Claude-1.3 performs well. For other models, performance degraded on this task.

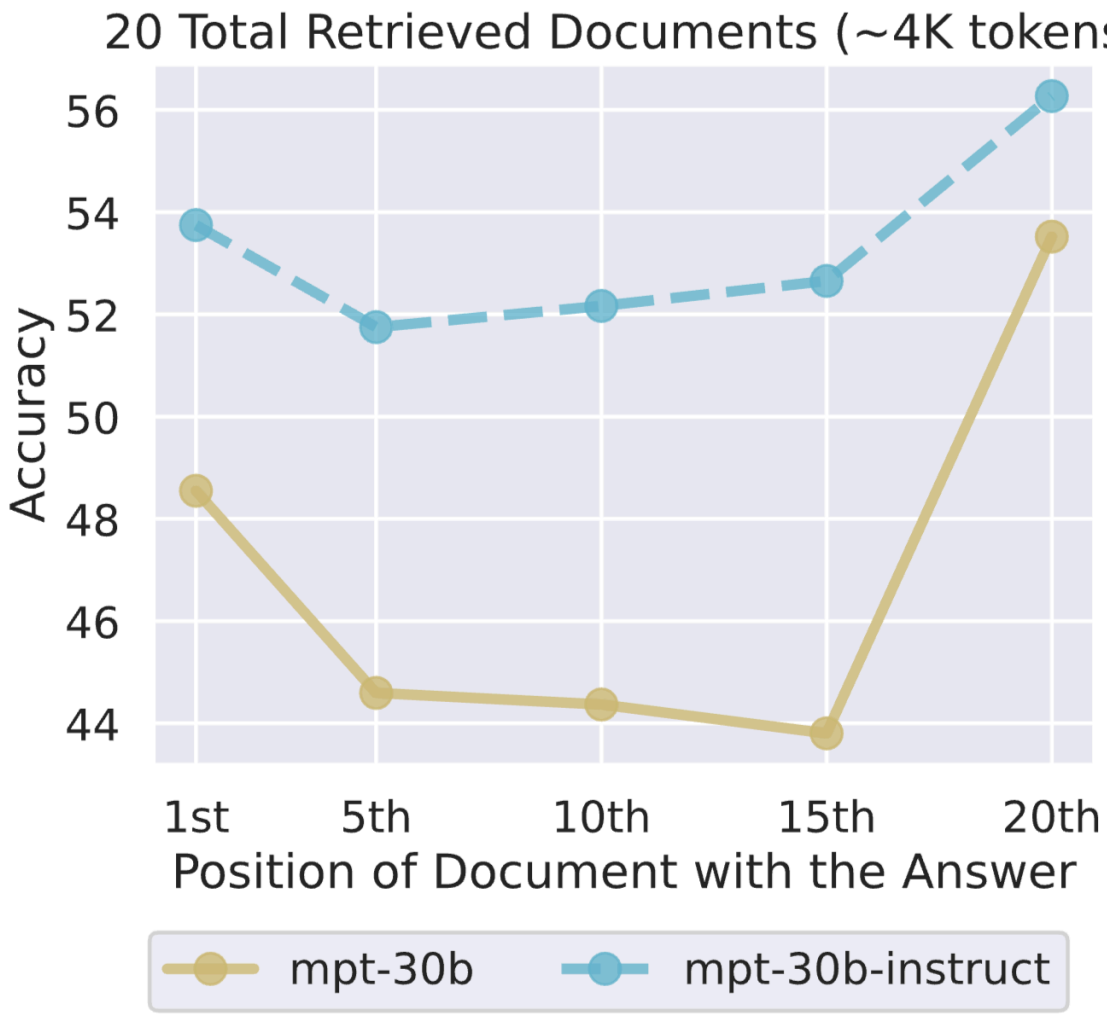


When encoder-decoder models are evaluated on input sequences shorter than the ones seen during training, robustness to change in information position is observed. On the other hand, if the input sequence is longer than the ones seen during training, the U-shaped curve is observed.

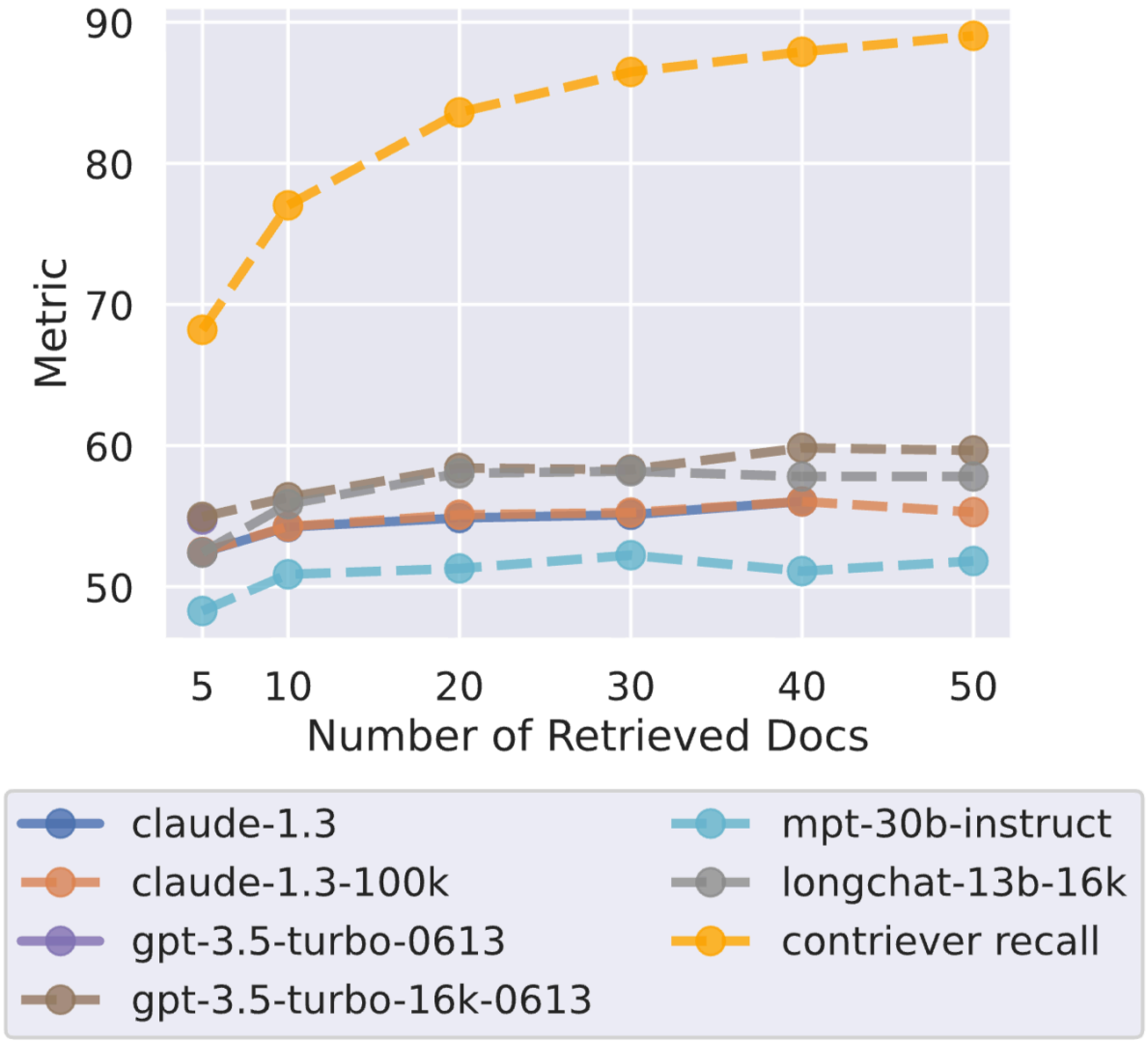


Placing the query before or after the input has no substantial effect in the model

performance. Performance slightly increases when relevant information placed at the beginning of the context.



Even models that are fine-tuned with instructions display a similar trend, although their absolute performance is higher.



Saturation of model performance, models face difficulty in using the retrieved information as the context grows.

Introduction

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.1

> hese models perform downstream tasks primarily via prompting: all relevant task specification and data to process is formatted as a textual input context, and the model returns a generated text completion

Lost in the middle only evaluates text modality.

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.1

> Existing language models are generally implemented with Transformers (Vaswani et al., 2017), which require memory and compute that increases quadratically in sequence length.

Every token attends to every other token

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.2

> n particular, our experiments make controlled changes to the input context size and the position of the relevant information within the input context and study their effects on language model performance

Authors do a controlled experimental analysis of the effect of the position of relevant information in the context window.

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.2

> We first experiment with multi-document question answering, which requires models to reason over provided documents to find relevant information and use it to answer a given question; this task mimics the retrieval-augmented generation setup underlying many commercial generative search and question answering applications (e.g., Bing Chat). In this setting, we control (i) the input context length by changing the number of documents in the input context (akin to retrieving more or less documents in retrieval-augmented generation), and (ii) control the position of the relevant information within the input context by changing the order of the documents to place the relevant document at

In multi-document question-answering, authors vary the number of input documents to control the sequence length and the position of relevant information. This provides the following insights:

1. How does the context length affect the model performance?
2. Is the model biased towards certain positions in the context which the authors describe as primary bias (beginning of sequence) and recency bias (end of sequence)?

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.2

> For example, when relevant information is placed in the middle of its input context, GPT-3.5-Turbo's performance on the multi-

> Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.2
>
> document question task is lower than its performance when predicting without any documents (i.e., the closed-book setting; 56.1%).

Basline method

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.2

> Given that language models struggle to retrieve and use relevant information in the multi-document question answering task, to what extent can language models even retrieve from their input contexts?

Third research question: to what extent can the models actually retrieve the information?

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> Encoder-decoder models are relatively robust to changes in the position of relevant information within their input context, but only when evaluated on sequences within its trainingtime sequence length.

Encoder-decoder models encode the entire sequence and employs cross-attention, thus enhancing its ability to attend to tokens within the context. For our experiment, we can also try to go beyond the context length the model see during training

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.2

> Query-aware contextualization (placing the query before and after the documents or keyvalue pairs) enables near-perfect performance on the synthetic key-value task, but minimally changes trends in multi-document QA

Idea for experimentation, placing the question at the beginning rather than at the end of the sequence. Can draw parallels with this paper and compare the recent models

with the newer ones.

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.3

> In contrast to our controlled multi-document QA task, where the context always contains exactly one document that answers the question, none or many of the top k documents may contain the answer in the open-domain QA setting.

In an open-domain setting, one/none/many documents can contain the answer. This can be analogous to out interleaved chat setting. The idea that one/none/many documents can contain the answer can also be used in our case, potentially.

Results and discussion

Multidocument QA

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> In particular, we make controlled changes to the length of the input context and the position of the relevant information and measure changes in task performance.

1. Setup: 1 document containing the answer, k-1 distractors. Distractors are related to the "key" document but do not contain the answer. Distractors are sorted in order of relevance. (Doesn't placing the answer between distractors cause breakage of the relevance order as the doc with the answer is actually the most relevant?)
2. Variables: Length of input context; position of information
3. Models used: MPT-30B-Int, Longchat-13B, LLaMA-13B, GPT-3.5-Turbo

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.5

> Model performance is highest when relevant information occurs at the beginning or end of its input context.

Finding 1

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.5

> Extended-context models are not necessarily better at using input context.

Finding 2

How good is the retrieval?

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.6

> Given that language models struggle to retrieve and use information from the middle of their input contexts in the multi-document question answering task, to what extent can they simply retrieve from input contexts?

1. Setup: Synthetic key-value retrieval task. K key-value pairs in JSON format and the question key are given as input. Model fetches the value corresponding to the question key.
2. Minimal natural language features - they can be confounders. Can we potentially add confounders to our experiment?
3. Models: same as above.

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.6

> laude-1.3 and Claude-1.3 (100K) do nearly perfectly on all evaluated input context lengths,

Finding 1

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.6

> Similar to our multi-document QA results, GPT- 3.5-Turbo, GPT-3.5-Turbo (16K), and MPT-30BInstruct have the lowest performance when they must access key-value pairs in the middle of their input context.

Finding 2

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.7

> Why Are Language Models Not Robust to Changes in the Position of Relevant

Why are language models not robust to changes in the position of relevant information?

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.7

> e hypothesize that encoder-decoder models may make better use of their context windows because their bidirectional encoder allows processing each document in the context of future documents, potentially improving relative importance estimation between documents.

But only within context window, for input beyond context window, performance degrades.

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.7

> decoder-only models cannot attend to query tokens when contextualizing documents or keyvalue pairs, since the query only appears at the end

Decoder only models attend to tokens occuring before, not in the future. Encoder-decoder models use a bi-directional encoder, helping them attend to both future and past context.

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.8

> We find that query-aware contextualization dramatically improves performance on the key-value retrieval task

When query is placed at the beginning, the performance improves for key-value task, but not as significantly in the multi-document task.

Lost in the Middle How Language Models Use Long Contexts (2307.03172v3), p.9

> To better understand the effect of additional finetuning and model scale, we also experimented with Llama-2 models of varying sizes (7B, 13B, and 70B) with and without additional supervised fine-tuning and reinforcement learning from human feedback (Appendix E). We find that the Ushaped performance curve only appears in sufficiently large language models (with or without additional fine-tuning)—the 7B Llama-2 models are solely recency biased, while the 13B and 70B models exhibit a U-shaped performance curve. In addition, we see that the Llama-2

> supervised fine-tuning and reinforcement learning from human feedback procedure slightly mitigates the positional bias in smaller models (13B, akin to trends shown when comparing MPT-30B and MPT-30B-Instruct), but minimally affects trends on larger models (70B).

1. Smaller models exhibit recency bias, larger models have U-shaped curves
2. Fine-tuning has limited impact.
3. Fine-tuning has minimal effect on positional bias in 70B models
4. Summary: Model scale significantly influences positional bias, whereas additional fine-tuning provides only modest improvements.

Is more context always better?

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> Our results indicate that prompting language models with longer input contexts is a trade-off— providing the language model with more information may help it perform the downstream task, but it also increases the amount of content that the model must reason over, potentially decreasing accuracy

1. more content in input, more content to reason over, lowers accuracy.
2. Reader model and retrieval model: reader model extracts the answer, retrieval model fetches documents.
3. Key finding: Retrieval model will keep fetching the documents, but reader model is not able to make good use of it and its performance saturates.